A Semantically Nonredundant Continuous-Scale Feature Network for Panchromatic and Multispectral Classification

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***Abstract*— In recent years, panchromatic (PAN) images and multispectral (MS) images, as a type of multimodal remote sensing data, are attracting increasingly more attention to their classification problems. However, effectively representing size variations of targets in remote sensing images and reducing redundant representations of different modalities' deep semantic features to enhance classification accuracy remains a challenge. In this article, we propose a semantically nonredundant continuous-scale feature network (SNCF-Net) for PAN and MS classification, consisting of two modules: the texture-enhanced continuous scale input generation module and the cross-modal feature Kernel interaction (CMKI) module. By simulating the human eye's adjustment of distance to observe objects of different sizes, we employ 3-D convolution to extract continuous-scale images generated by the texture-enhanced continuous-scale input generation (TCIG) module, enabling optimal feature representation of objects in remote sensing images. Additionally, the texture enhancement (TE) strategy in the TCIG module alleviates texture diffusion in scale space, enhancing the network's ability to represent texture features. Subsequently, the CMKI module utilizes the response differences between different features to generate convolution kernels from deep feature maps, enabling feature interaction between the PAN modal and MS modal. This reduces redundant representations of essential image content information in deep features of two modalities, facilitating a better mapping between dual-modal features and categories. Our results achieve state-of-the-art performance on multiple datasets. The code is available at https://github.com/Xidian-AIGroup190726/SNCFNet.**

***Index Terms*— Continuous scale, deep learning, feature interaction, multisource images classification, panchromatic (PAN) and multispectral (MS), texture-enhanced.**

I. Introduction

IN RECENT years, there has been an increasing interest in the fusion of different remote sensing data sources to improve image classification accuracy [1], [2], [3]. In particular, the fusion of panchromatic (PAN) and multispectral (MS) data has received considerable attention due to the complementary information provided by these two data sources [4].

PAN data have high spatial resolution but low spectral resolution, while MS data have low spatial resolution but high spectral resolution [5], [6]. Therefore, the fusion of PAN and MS data can take advantage of both the high spatial resolution of PAN data and the high spectral resolution of MS data to produce a high-resolution MS image, which is more suitable for image classification.

Research on remote sensing data fusion methods can generally be classified into two categories: data-level fusion [7], [8], which involves designing methods to preserve the advantageous characteristics of various data sources and produce a fused image that incorporates these diverse advantages. Feature-level fusion [9], [10], on the other hand, is primarily aimed at supporting downstream tasks such as classification, segmentation, and detection. It involves integrating features extracted from different data sources to enhance task performance.

For PAN and MS data, the former category aims to transfer the rich spectral information from the MS image to the high spatial resolution of the PAN image, resulting in a high-resolution MS image. The key aspect of this category of methods lies in the pan-sharpening algorithm [11]. Recently, various excellent pan-sharpening algorithms have been proposed. Ma et al. [5] proposed an unsupervised pan-sharpening method based on generative adversarial networks. They designed loss functions that encourage the generated image to have spectral information similar to the MS image while maintaining spatial details similar to the PAN image. Zhou et al. [12] also proposed a method based on distillation learning, using a teacher model to extract image features and a student model to perform pan-sharpening. However, the fused high-resolution MS image often contains a significant amount of redundant information, which can adversely affect downstream classification tasks.

With the rapid development of deep learning [13], [14], [15], the latter approach of feature extraction followed by classification has become a hot research topic. Zhu et al. [16] proposed a dual-branch attention fusion network that highlights the advantageous features of different data sources, but did not consider redundancy between them. Liao et al. [17] proposed

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TABLE I
COMPARATIVE ANALYSIS OF EXISTING METHODS

Methods	Advantages	Disadvantages
DBAF-Net [16]	Simple modular structure, plug and play.	Not considering redundancy in dual-source features.
TSMF-Net [17]	Feature fusion stage fully considers redundant features.	Data fusion stage compromises the advantageous characteristics of the image.
MFT [18]	Applicable to various remote sensing images.	Not considering the characteristics and feature redundancy of different images.
AsyFFNet [19]	Parameter sharing, flexible structure.	The nature of the features is unclear, with poor interpretability.
DPCL [20]	Hierarchical training results in faster network convergence.	Excessive feature stacking and addition result in a bloated network structure.

a two-stage mutual fusion network that reduces modality differences at both the data and feature levels. Roy et al. [18] designed a transformer architecture network to fuse information from multimodal images for land cover classification. But they did not fully consider the advantageous characteristics of different images. Li et al. [19] eliminated unnecessary channels and enhance the network's generalization capability by measuring the redundancy of different channels using a scaling factor. Zhu et al. [20] designed an adaptive neighborhood sample grading strategy and simulated a biologically inspired design loss function to train the network. This approach effectively modeled the object relationships between different layers, allowing for more comprehensive information integration. We briefly summarized the advantages and disadvantages of these methods, as detailed in Table I.

The mentioned methods have improved classification accuracy by operating on both the data and feature levels of multimodal remote sensing images. The multiscale approach [21], [22], [23] is a common technique in this field, which involves partitioning objects of different sizes in an image into specific scales to facilitate the extraction of features for objects of varying sizes. Sun et al. [24] used a multiscale supervised training approach to constrain the outputs of the multimodal decoder to different sizes of images. Ding et al. [3] achieved the extraction of local features at different scales by adjusting the size of input images, followed by cascading and processing features from different modalities for classification.

However, in reality, the sizes of objects in remote sensing images vary continuously, and extracting features at only a few scales makes it difficult to obtain a good representation for all objects. Simultaneously, it is also a challenge to better extract edge and texture features in images that are essential to classification across multiple scales. Additionally, although PAN and MS data have distinct characteristics, they still represent the same essence of the area. This implies that besides the unique features originating from the characteristics of each data type, there exists a considerable portion of shared features within both data types. Both their shared features and their individual unique features exhibit different representations on various samples and different categories of land objects. Therefore, all of these features contribute positively to improving classification performance. To establish an excellent classifier, it is necessary to reduce the redundant representation of shared features between the dual-source features inputted into the classifier.

Based on these considerations, this article proposes a semantically nonredundant continuous-scale feature network (SNCF-Net). Our contributions and innovations can be summarized as follows.

1) We propose a texture-enhanced continuous-scale input generation (TCIG) module, which provides a basis for network simulation of the way the human eye observes objects of different sizes. TCIG establishes a continuous-scale space in which the network can perceive objects of changing size, so that all features in the image can be observed at the optimal scale to obtain better data feature representation. The texture enhancement (TE) strategy in TCIG alleviates the texture diffusion phenomenon in scale space and improves the characterization ability of the network for image texture features.
2) We propose a cross-modal feature Kernel interaction (CMKI) module, which interacts with features from two modalities by generating convolution kernels to reduce redundant representations of shared features in dual-modal deep features. We use deep feature maps to generate convolution kernels with directivity and orthogonality, which filter the shallow features of the other modality. By exploiting response differences between features with different properties, we suppress their unique characteristics and obtain shared features between the two modalities. The original deep features still contain a complete representation of both shared and unique features. We eliminate the shared features from them, obtaining unique features, and then fuse the shared features from both modalities to reduce redundancy in the input to the classifier.
3) We propose a SNCF-Net for PAN and MS classification. This network progressively blends the continuous-scale images generated by TCIG to obtain different continuous-scale blended features. Each channel of these features is three-dimensionally continuous (length, width, and scale). Based on this, we use CMKI to reduce redundancy in shared features within the deep layers of the two modalities, achieving better feature representation for dual-modal data. This allows us to build an excellent classifier and achieve state-of-the-art results on four different datasets.

The rest of the article is organized as follows. Section II provides the preliminaries to this article's method; Section III

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describes the proposed method in detail; Section IV presents the experimental results and analysis; and finally, Section V concludes the article and discusses future work.

II. Preliminaries

A. Scale Space

Scale space is an attempt to simulate the concept and method of human eyes observing objects in the image field [25]. The basic concept of scale space theory is that structures and features in images can be described and analyzed at different scales [26]. The scale of the image can be understood as the change of the viewing angle of the observed object or the smoothness of the image. Coarse scales can be used to detect large-scale structures and overall shapes, while finer scales can be used to detect small-scale details and edges. By analyzing images at different scales, multiscale features such as edges, corners, and textures, can be extracted. These features are useful for tasks such as object detection, image matching, and classification.

The scale space of an image refers to the degree of blurring of the image, not the size of the image. Smoothing can be achieved using a Gaussian filter, where the scale of the filter determines the degree of smoothing. It is worth mentioning that Babaud et al. [27] have proved that the Gaussian kernel is the only transformation kernel to achieve scale transformation through different approaches in precise mathematical forms. Given an image function $I(x, y)$, where x and y are spatial coordinates, a Gaussian scale space can be constructed by a series of Gaussian filters. Each filter has a specific scale parameter, usually denoted by a standard deviation σ. In the Gaussian scale space, by applying a series of Gaussian filters of different scales to the original image $I(x, y)$, a series of smooth images can be obtained, each corresponding to a specific scale. These smooth images can be expressed as: $L(x, y; \sigma)$. By changing the scale parameter σ, different degrees of smoothing effects can be obtained at different scales. Smaller σ values correspond to finer scales, which can extract detailed information, while larger σ values correspond to coarser scales, which can capture a wider range of structures.

B. Wavelet Transform

Wavelet transform [28] can represent a signal or image as a linear combination of wavelet basis functions of different scales and frequencies. Wavelet transform has the characteristic of time-frequency localization, and is very useful for capturing instantaneous features and local details in signals or images. Wavelet transform can be applied to multidimensional data. When using it to process 3-D data, it can be expressed as

$$W(a, b, c) = \sum_{x=0}^{X-1} \sum_{y=0}^{Y-1} \sum_{z=0}^{Z-1} V(x, y, z) \cdot \psi_{a,b,c}(x, y, z) \quad (1)$$

where, $W(a, b, c)$ represents the wavelet coefficient, $V(x, y, z)$ is the original signal or image, and $\psi_{a,b,c}(x, y, z)$ is the wavelet basis function. Wavelet basis functions have the characteristics of finite length and compact support, and are usually constructed on the basis of mother wavelet functions. There are many choices of common wavelet basis functions, such as Haar wavelet, Daubechies wavelet, and Symlet wavelet. They have different local properties in frequency and time domain, and are suitable for different application scenarios.

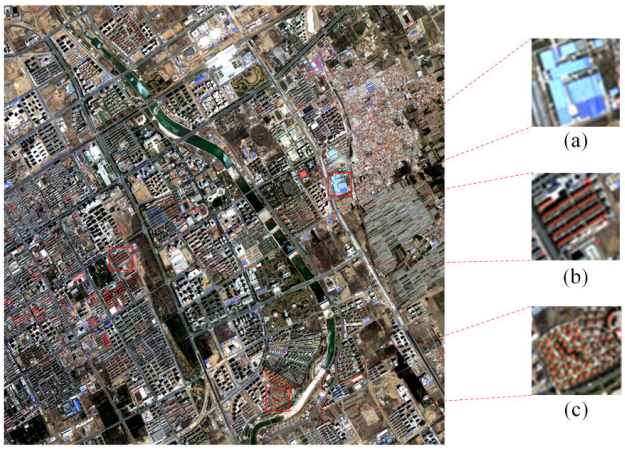


Fig. 1. Difference of the scale of different ground objects. (a) and (b) Different categories of objects, with significant differences in scale and direction. (b) and (c) Belong to the same category of objects, yet their geometric properties also exhibit considerable differences.

III. Method

In this section, we introduce the proposed SNCF-Net. There are two designed components of the SNCF-Net, including the TCIG and CMKI.

A. Texture-Enhanced Continuous-Scale Input Generation

1) Necessity of Continuous-Scale: As shown in Fig. 1, remote sensing images contain ground objects of different categories with significant differences in size and diverse orientations. Even ground objects of the same category often exhibit various geometric properties. Simple neural networks struggle to fully model the geometric properties of different ground objects in remote sensing images, which hinders achieving high-quality feature extraction to enhance the performance of downstream classification tasks [29]. Multiscale methods can effectively address this issue. However, existing multiscale methods [30], [31] often perform feature extraction of ground objects with different geometric properties (large, medium, and small ground objects) on different levels of feature maps. This is achieved by adjusting convolution strides or pooling operations to achieve upsampling or downsampling of feature maps, extracting features of ground objects with different geometric properties on feature maps of certain discrete scales.

In reality, the geometric characteristics of different ground objects are constantly changing, and there is no established standard that can perfectly distinguish them into discrete scales for feature extraction. Moreover, the human eyes can adjust and observe objects at the optimal observation scale by modifying the distance to the observed objects within a

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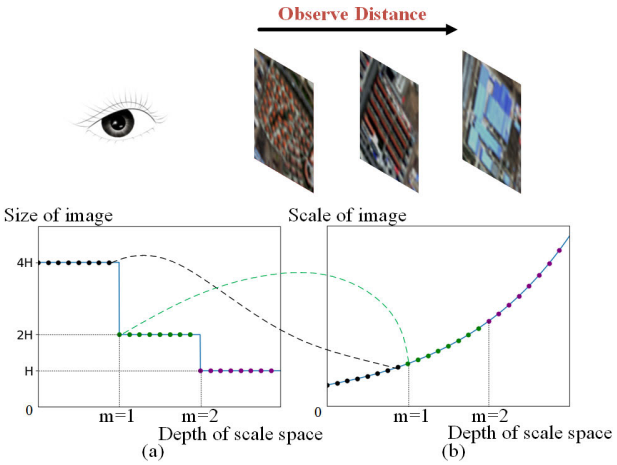


Fig. 2. Illustration of image size and scale in a Gaussian pyramid. (a) Depicts the relationship between image size and the depth of scale space in the pyramid. (b) Illustrates how the scale of images in the pyramid gradually increases with the deepening of scale space. Both aspects reflect that image size is discrete within the pyramid, while scale is continuous.

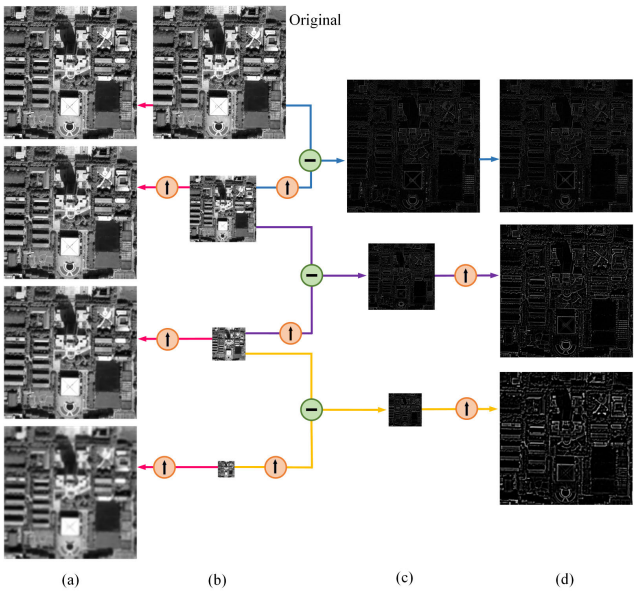


Fig. 3. Depiction of the changing texture information in the Gaussian blur process. In the (b) column, the first image is the original picture, and the images below it are progressively blurred. In the (a) column, the images from the (b) column are magnified to a uniform size. In the (c) column, the lost information between adjacent levels of images in the (b) column is shown. In the (d) column, the images from the (c) column are magnified to the same size.

continuous range. We want computers to have the similar ability, where they can dynamically adjust their observation scale to perceive objects accurately.

2) Generation of Continuous-Scale Images: To address the issues mentioned above, we attempt to make the network simulate the process of the human eyes searching for the optimal distance to observe ground objects with different geometric properties within a continuous range. We aim for the network to traverse the spatial features of the image in 2-D space while simultaneously perceiving ground objects with different geometric properties in the scale space. This allows the network adaptively select the appropriate scale to extract features of ground objects in the image and utilize these extracted features for classification. To achieve this, we need to obtain images at continuous scales. Images of different scales can be obtained by Gaussian blurring, that is, using a Gaussian filter to convolve the input image

$$L(x, y, \sigma) = G(x, y, \sigma) \otimes I(x, y) \tag{2}$$

where $I(x, y)$ is the image needed to be blurred, $G(x, y, \sigma)$ is the Gaussian function

$$G(x, y, \sigma) = \frac{1}{2\pi\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}} \tag{3}$$

where σ is the blur factor, and the larger the σ value is, the more blurring the image is.

The Gaussian pyramid uses Gaussian blur to generate m groups of n layers, with a total of $m \times n$ images at continuous scales. Among them, the size of the images in the ith group is twice that of the $(i + 1)$th group, and the scale of images within each group varies uniformly. The exponential function with base 2 is used to achieve continuous scale between different groups of images, as shown in Fig. 2. When the scale difference between adjacent images is small enough, we can consider these images to be continuous-scale. The blur factor of the ith group kth layer image in the Gaussian pyramid can be calculated by the following formula:

$$\sigma_{i,k} = \sigma_0 \cdot 2^{i+k/n} \tag{4}$$

where $i \in (0, 1, 2, \ldots, m-1)$, $k \in (0, 1, 2, \ldots, n-1)$. σ_0 is a hyperparameter.

3) Implementation of TE: In addition to the differences in target sizes, remote sensing images also contain abundant multidirectional texture information, reflecting the texture variations between different regions. This plays a crucial role in the classification results. However, during the process of generating continuous-scale images through Gaussian blurring and downsampling, the image loses a portion of its texture information. As shown in Fig. 3, the differences between adjacent levels of the Gaussian pyramid are referred to as the Laplacian pyramid (LP). In the (d) column images of Fig. 3, the texture information becomes coarser, indicating that the texture information in the image spreads around during the Gaussian blurring process. This increases the difficulty for neural networks to extract texture features and ultimately impacts the classification results.

To preserve the image's texture information while expanding the image scale, we need to perform texture feature enhancement during the continuous Gaussian blurring process. However, as the image scale increases, dense and cluttered texture information within the image should gradually disappear. To maintain the representation of image features in a continuous-scale space, we only enhance the texture information along several prominent directions in the image, while the dense and cluttered texture information within regions is still suppressed by Gaussian blurring. This approach allows us to obtain images with distinct regional boundaries and gradually consistent textures within each region in the continuous scale space. Therefore, we need to extract the texture information from the image in several different directions, and the contourlet transform is a local and directional image

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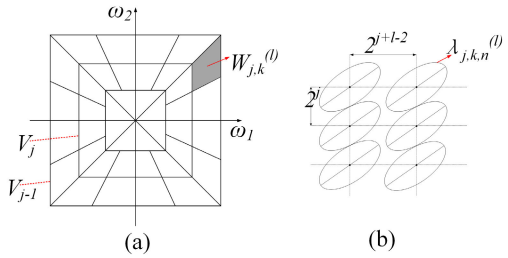


Fig. 4. Contourlet subspaces. (a) Frequency domain partitioning of multiscale, multidirection subspaces achieved through LP and DFB. (b) Sampling grid and approximate support of contourlet functions for a "mostly horizontal" subspace $W_{j,k}^{(l)}$.

representation method [32] has the potential to address this task.

The contourlet transform is a dual-filter bank structure composed of LP and directional filter banks (DFBs). As mentioned above, the LP is the difference between adjacent 2^j resolution images V_j and V_{j-1}, as illustrated in Fig. 4

$$V_{j-1} = V_j \oplus W_j. \tag{5}$$

In other words, the contourlet transform divides W_j using DFB into multiple directional subbands. The DFB are realized through a hierarchical binary tree decomposition of level l, segmenting the bandpass image W_j into several directional wedges, thereby providing directional information

$$W_j = \bigoplus_{k=0}^{2^l-1} W_{j,k}^{(l)} \tag{6}$$

where the j and k specify the scale and direction, respectively. The image can be decomposed using an equivalent parallel-structure filter bank, and the response sampling matrix takes the following diagonal form:

$$S_k^{(l)} = \begin{cases} \text{diag}\left(2^{l-1}, 2\right), & \text{for } 0 \le k < 2^{l-1} \\ \text{diag}\left(2, 2^{l-1}\right), & \text{for } 2^{l-1} \le k < 2^l. \end{cases} \tag{7}$$

The two components in the above equation correspond, respectively, to the majority of the horizontal direction and the majority of the vertical direction. Finally, the representation of the subspace $W_{j,k}^{(l)}$ on the tight frame $\{\lambda_{j,k,n}^{(l)}(t)\}_{n\in Z^2}$ [32] is realized by applying the DFB, $d_k^{(l)} \sim F_{\text{DFB}}$

$$\lambda_{j,k,n}^{(l)}(t) = \sum_{m\in Z^2} d_k^{(l)}\left[m - S_k^{(l)} n\right] W_j. \tag{8}$$

See Fig. 4(b), the subspace $W_{j,k}^{(l)}$ is defined on a rectangular grid with intervals $2^{j+l-2} \times 2^j$ or $2^j \times 2^{j+l-2}$, depending on whether it is mostly horizontal or vertical. The frame itself is orthogonal, and the partitioning of directional subspaces is also orthogonal. Therefore, the contourlet transform can orthogonally extract texture information from multiple directions. Based on this, we designed a TCIG module for PAN and MS classification, as illustrated in Fig. 5.

The input image first goes through the LP to obtain multiscale low-frequency and high-frequency components. The high-frequency components then pass through the DFB to obtain the image's contours in different directions. The upsampled PAN image is transformed by contourlet to obtain multidirectional texture information corresponding to each group size of the Gaussian pyramid

$$X_{\text{low},i}, X_{\text{high},i} = I_{\text{pan}}(x, y) \uparrow \otimes F_{\text{LP}} \tag{9}$$

$$X_{i,j} = X_{\text{high},i} \otimes F_{\text{DFB}} \tag{10}$$

where F_{LP} and F_{DFB} are the LP and DFB; i represents the levels of contourlet transform, which can correspond to the group number of Gaussian pyramid; j represents the direction of the texture information extracted from the picture. The subscripts low and high of X represent the low- and high-pass components, respectively. We use texture information from different directions $X_{i,j}$ to enhance the edge texture features of the image, mitigating the impact of Gaussian blur on these prominent directions' texture features

$$L_{\text{pan}}\left(x, y, \sigma_{i,k}\right) = G\left(x, y, \sigma_{i,k}\right) \otimes \left[L_{\text{pan}}\left(x, y, \sigma_{i,0}\right) + \sum_{j=0}^{n} X_{i,j}\right]. \tag{11}$$

We add the extracted texture information of different levels of PAN to the 0th image of the corresponding group in the Gaussian pyramid, and then continue to perform Gaussian blurring to obtain continuous-scale images with enhanced texture information. The $L_{\text{pan}}(x, y, \sigma_{i,k})$ represents the ith group and the kth layer image of PAN.

For MS image, the texture information in the PAN image is more accurate. So, we add the extracted PAN texture information to the all channels of MS, and each channel independently generates a Gaussian pyramid. This operation not only helps alleviate the texture diffusion phenomenon but also enhances MS's spatial features. Similarly, the ith group kth image generated by the c channel of MS can be expressed as

$$L_{\text{ms}}\left(x_c, y_c, \sigma_{i,k}\right) = G\left(x, y, \sigma_{i,k}\right) \otimes \left[L_{\text{ms}}\left(x_c, y_c, \sigma_{i,0}\right) + \sum_{j=0}^{n} X_{i,j}\right]. \tag{12}$$

In fact, in order to maintain scale continuity between different groups of images, the 0th image of the next group in the Gaussian pyramid is obtained by downsampling the last image of the previous group after Gaussian filtering

$$L\left(x, y, \sigma_{i+1,0}\right) = G\left(x, y, \sigma_{i,n}\right) \otimes \left(L\left(x, y, \sigma_{i,n-1}\right)\right) \downarrow. \tag{13}$$

In this way, we obtain continuous-scale images enhanced by PAN and MS texture information, respectively. While blurring the image regions, we reduced the loss of texture information, preserving the texture differences between different regions of the image.

B. Cross-Modal Feature Kernel Interaction

1) Feature Extraction From Dual-Source Continuous-Scale Images: Although PAN and MS data have different characteristics, they still represent the same underlying essence of the same region. Therefore, in addition to their respective advantageous features, they also exhibit many similarities in

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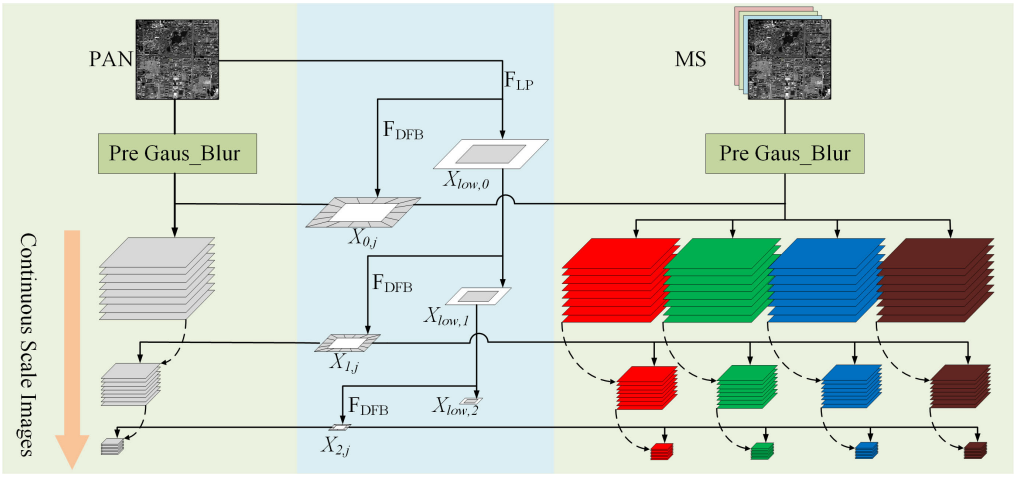


Fig. 5. Proposed TCIG. Compared with MS images, PAN images have better representation of spatial structure information. Therefore, we use the multilevel contourlet transform to extract the multidirectional texture information of the PAN image, and enhance the texture features of the PAN and MS images when generating the Gaussian pyramid.

deep semantic features. For classifiers, redundant similar features between the two branch features are detrimental because they can affect the classifier's ability to establish a good mapping between the advantageous features of dual-source data and their corresponding categories. Therefore, designing a method to extract shared features that represent their common essence and unique features that represent their advantageous characteristics from dual-source data is crucial for our task.

Through the TCIG module, we obtain continuous-scale images of both PAN and MS data. We utilize neural networks to extract their features for image classification. In order to maintain the scale continuity of feature maps within the network, we employ 3-D convolutions to process the continuous-scale images generated in our approach. The network traverses the scale space, extracting spatial features of the images at continuous scales. As the network deepens, these features are aggregated into semantic features for image classification. In 2-D convolution networks, feature maps lack explicit meaning in the feature channel dimension. Different from the 2-D convolution network, we introduce an extra dimension representing the scale for the feature map, and use 3-D convolution to process them, so that each channel of the feature map in the network is 3-D continuous. This extra continuous dimension opens up more possibilities for us to operate on the feature maps.

2) Generating Convolution Kernels With Directivity and Orthogonality: Thanks to the 3-D continuity of feature maps in each channel, we use deep feature maps of dual-source data to generate convolution kernels for extracting shared features. As shown in Fig. 6, we utilize deep semantic features from two branches to generate convolution kernels that possess directivity and are mutually orthogonal. The 3-D discrete wavelet transform (DWT) is used to losslessly extract the information of the deep feature map, while compressing the size of the feature map to generate a smaller convolution kernel. For each input 3-D feature map, eight components (LLL, LLH, LHL, LHH, HLL, HLH, HHL, HHH) with distinct directional attributes will be generated after wavelet transformation

$$\text{LLL, LLH, LHL, } \ldots \text{, HHH} = \text{3D DWT}\left(f^{\text{deep}}\right) \tag{14}$$

where f^{deep} is the deep semantic features of feature extraction branches. Take the PAN branch as an example, then, the eight components are concatenated together for statistical generation of convolution kernels

$$\text{kernel}_{\text{pan}} = \text{Concat(LLL, LLH, LHL, } \ldots \text{, HHH)}. \tag{15}$$

For the input of $[N, C, S, H, W]$, the output dimension of the above formula is $[N, 8 \times C, S/2, H/2, W/2]$. The generated convolution kernel is applied to a convolution layer, where C_{in} is the input channel of the convolution layer, and C_{out} is the output channel, we finally get the output of the $[C_{\text{in}}, C_{\text{out}}, S/2, H/2, W/2]$ dimension. Therefore, we need to perform data statistics on the convolution kernel of $8 \times C$ components of N samples, corresponding to the number of input and output channels of the upper convolution layer.

To preserve the spatial distribution of features to the maximum extent possible, we use $1 \times 1 \times 1$ group convolution to map $8 \times C$ component channels to C_{out} channels

$$\text{kernel}_{\text{pan}} = \text{Conv}_{\text{grpup}}(\text{kernel}). \tag{16}$$

The number of convolution integral groups is eight, corresponding to the eight components extracted by wavelet transform. In this way, the direction of the wavelet transform output components can be preserved to the greatest extent, so that the generated convolution kernels have their own characteristics, and the redundancy of the output of the convolution layer is reduced. We average the statistics of different samples N(Batchsize) to C_{in} channels, and the number of statistical samples on each channel of C_{in} is calculated by the following formula:

$$N_{\text{samp}} = \text{Batchsize}/C_{\text{in}}. \tag{17}$$

The feature distribution of different samples statistically grouped by C_{in} is concatenated together to obtain

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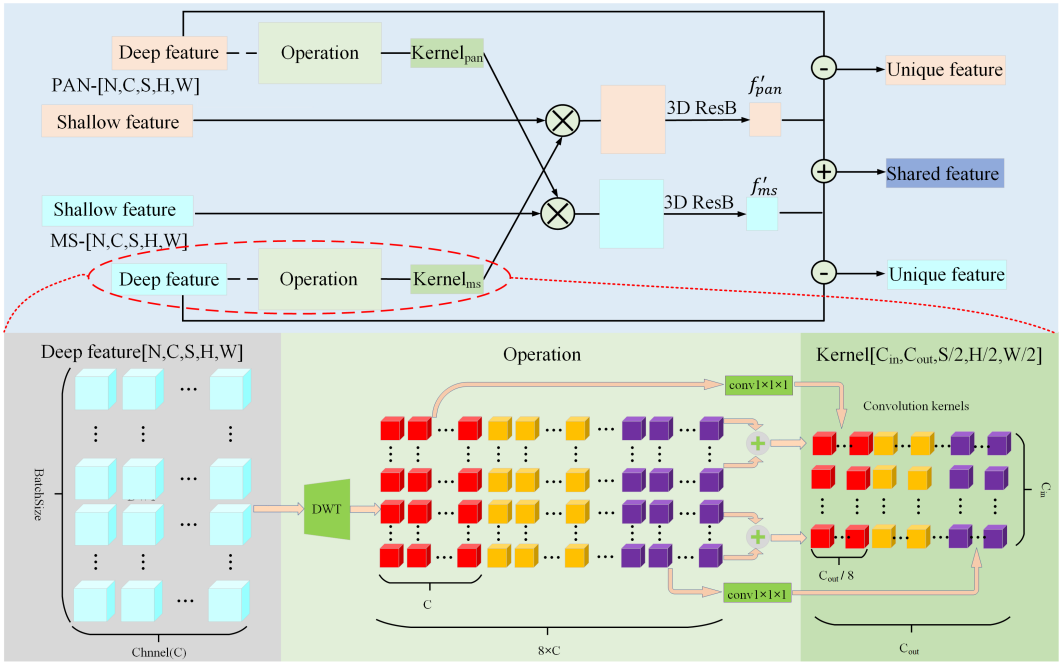


Fig. 6. Proposed CMKI. CMKI uses deep features to generate directional and orthogonal convolution kernels and utilizes these kernels to filter shallow features from the other modality, resulting in shared features between the two modalities. The original branch's deep features minus the shared features give rise to modality-unique features.

a convolution kernel with the final dimension $[C_{\text{in}}, C_{\text{out}}, S/2, H/2, W/2]$

$$\text{kernel}_{\text{pan}} = \text{Concat}\left(\sum_{i\cdot N_{\text{samp}}}^{(i+1)N_{\text{samp}}} \text{kernel}_{\text{pan}}[i]\right) \quad i \in (0, 1, \ldots, C_{\text{in}} - 1). \tag{18}$$

Similarly, we can get the convolution kernel $\text{kernel}_{\text{ms}}$ generated by the MS branch. In this way, by utilizing the deep semantic feature maps to generate convolution kernels, we can use these convolution kernels to extract features. It is worth mentioning that the above process of generating the convolution kernel, including the wavelet transform used [33], is learnable.

3) Separation of Shared and Unique Features: We use the convolution kernel generated by the deep feature map of the PAN branch to extract features from the shallow feature map of the MS branch. Similarly, we use the convolution kernel generated by the deep feature map of the MS branch to perform feature extraction on the shallow feature map of the PAN branch

$$\begin{aligned} f'_{\text{pan}} &= f^{\text{shallow}}_{\text{pan}} \otimes \text{Kernel}_{\text{ms}} \\ f'_{\text{ms}} &= f^{\text{shallow}}_{\text{ms}} \otimes \text{Kernel}_{\text{pan}}. \end{aligned} \tag{19}$$

From a signal processing perspective, responses between signals with similar feature distributions are stronger compared to those between signals with different feature distributions [34], [35]. In other words, the response of the shared part between the deep semantic features of dual-source data are stronger than the response of the unique parts. Based on this, we employ a deep semantic feature map of one dataset to generate convolution kernels for filtering the shallow features of another dataset to extract shared and unique features from dual-source data. Due to the differences in feature responses between the two datasets, the response of unique features is effectively suppressed, while the response of shared features is effectively preserved. By leveraging the differences in response between different attribute features and these convolution kernels, we can suppress unique features and obtain shared features. This process allows us to extract shared features between two branches, and we further extract features using 3-D residual blocks from them to obtain f'_{pan} and f'_{ms}.

The f'_{pan} and f'_{ms} suppress unique features, highlighting shared characteristics. Meanwhile, the input deep features encompass all attributes of their corresponding branches. By subtracting these features from the shared features of that branch, we extract the unique features

$$\begin{aligned} \text{Unique}_{\text{pan}} &= f^{\text{deep}}_{\text{pan}} - f'_{\text{pan}} \\ \text{Unique}_{\text{ms}} &= f^{\text{deep}}_{\text{ms}} - f'_{\text{ms}}. \end{aligned} \tag{20}$$

The shared features of the two reflect the homogeneous features of the same area, so averaging the $f'_{\text{pan}}, f'_{\text{ms}}$ is the final shared feature

$$\text{Shared} = 1/2\left(f'_{\text{pan}} + f'_{\text{ms}}\right). \tag{21}$$

Through the deep and shallow feature interaction between the dual-source data, we obtain the nonredundant feature representations of the two types of data. This forms the foundation for constructing an effective classifier.

C. Semantically Nonredundant Continuous-Scale Feature Network

In this section, we introduce the proposed SNCF-Net in detail, as shown in Fig. 7. The SNCF-Net utilizes the two modules proposed above, so that the feature maps in the

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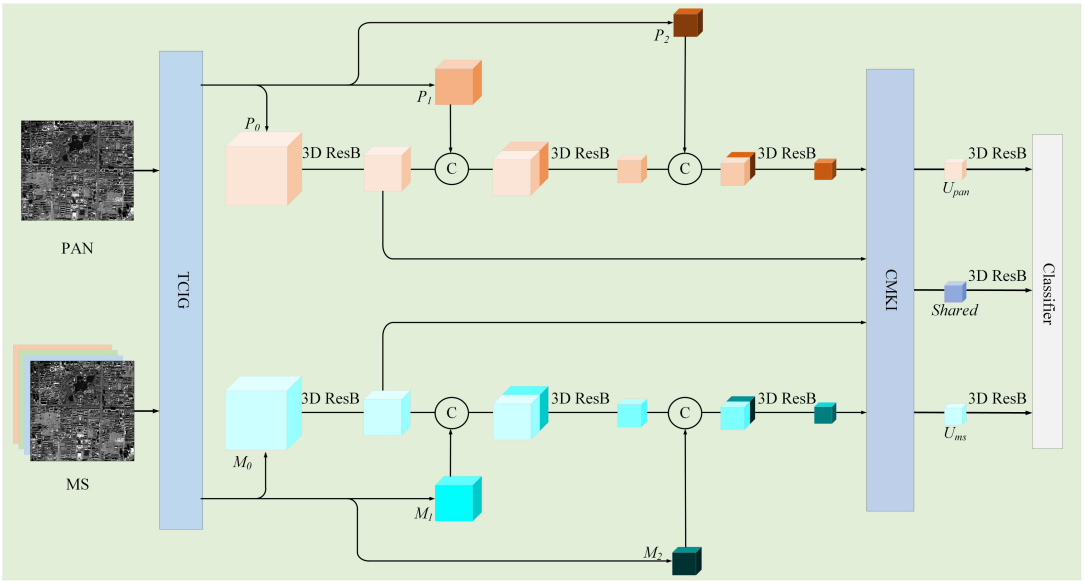


Fig. 7. Network structure of the proposed SNCF-Net. The figure depicts the variation of the feature map in a network using a single-channel feature map as an example. The feature map concatenation is done along the scale direction.

network have not only 2-D spatial continuity, but also 3-D continuity in the scale direction. In addition, since the 3-D feature maps of different channels are extracted by different convolution kernels, the continuous scale distribution of each 3-D feature map is also different.

The input of the network is the patch generated by PAN and MS image slices. The size of the patch generated by the PAN image is four times that of the MS image patch. We first upsample the MS patch to the same size of the PAN patch, then input it into the network. After the TCIG, the PAN and MS images, respectively, output m groups continuous scale images, each of which has n scale images

$$P_0, P_1, P_2, M_0, M_1, M_2 = \mathrm{CSIG}(\mathrm{PAN}, \mathrm{MS}) \tag{22}$$

where P_0 and M_0 is the input of the PAN branch and the MS branch, P_1, P_2, M_1, M_2 is embedded into two branches step by step. We use $1 \times 1 \times 1$ conv to make them have the same channels as the feature maps they will be embedded in

$$\begin{aligned} P_i' &= \mathrm{Relu}(Bn(\mathrm{Conv}(P_i))) \\ M_i' &= \mathrm{Relu}(Bn(\mathrm{Conv}(M_i))) \end{aligned} \tag{23}$$

then concatenate them with feature maps which have the same size as them

$$\begin{aligned} {f_{\mathrm{pan}}^i}' &= \mathrm{Concat}\left(f_{\mathrm{pan}}^i, P_i'\right) \\ {f_{\mathrm{ms}}^i}' &= \mathrm{Concat}\left(f_{\mathrm{ms}}^i, M_i'\right) \end{aligned} \tag{24}$$

where f_{pan}^i is the output of the ith 3-D Res_Block in the PAN branch, and f_{ms}^i is the output of the ith 3-D Res_Block in the MS branch. The output features of the first residual block and the third residual block in both branches are taken as the input shallow features and deep features for CMKI, respectively,

$$U_{\mathrm{pan}}, U_{\mathrm{ms}}, \mathrm{Shared} = \mathrm{CMKI}\left(f_{\mathrm{pan}}^1, f_{\mathrm{pan}}^3, f_{\mathrm{ms}}^1, f_{\mathrm{ms}}^3\right). \tag{25}$$

The shared features of PAN and MS, as well as the unique features of both have their own influence on the classification results. Therefore, we pass the U_{pan}, U_{ms}, and Shared through a 3-D Res_Block, perform average pooling, and concatenate them together, and input them into the fully connected layer. The loss function for classification is the cross-entropy function

$$\mathrm{loss} = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log\left(\hat{y}_i\right) + (1 - y_i)\log\left(1 - \hat{y}_i\right)\right]. \tag{26}$$

IV. Experiments

We introduce the datasets we used below. Afterward, there are experimental settings and the final experimental results that demonstrate the effectiveness of our proposed method.

A. Dataset Description

We conducted experimental analysis on our proposed method using four datasets. Each dataset consists of PAN and MS captured by the GF-2 satellite, along with ground truth obtained through annotation of the MS.

1) Hohhot Dataset: As shown in Fig. 8(a), the Hohhot dataset contains a total of 11 manually annotated classes (from GT1 to GT11, they are as follows: Water, Road, Bare Land, Vegetation, Intensive Building, Sparse Building, Suburb Building, Farmland, White Building, Red Building, and Blue Building). The MS consists of four channels, including red, green, blue, and near-infrared, with a resolution of $2001 \times 2001 \times 4$. On the other hand, the PAN has only one channel with a resolution of $8004 \times 8004 \times 1$. Among these classes, the GT5 has the largest number of train and test samples, accounting for 2159 train samples and 50 252 test samples.

2) Xi'an Dataset: As shown in Fig. 8(b), the Xi'an dataset contains a total of 12 classes (from GT1 to GT12, they are as follows: Water, Road1, Road2, Bare Land, Low Vegetation, Farmland, Tree, Intensive Building, Sparse Building, Suburb Building, Workshop Building, and White Building). The MS consisting of $4548 \times 4544 \times 4$ pixels. The PAN contains

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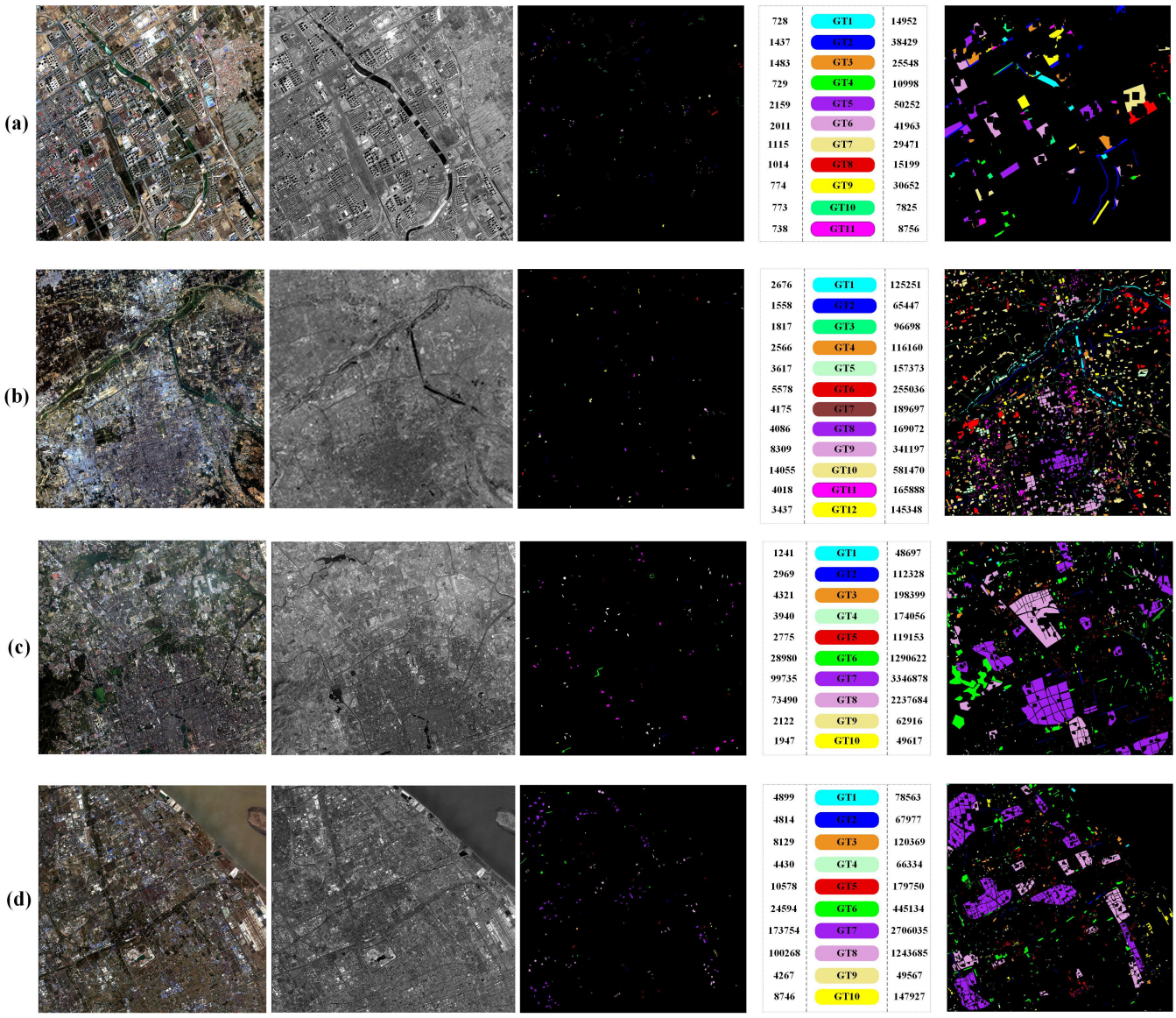


Fig. 8. Dataset used. The first column consists of MS images, the second column consists of PAN images, the third column contains the train set, the next is the number of each class in the datasets, and the last column is the ground truth of the test set. (a) Hohhot dataset, (b) Xi'an dataset, (c) Beijing dataset, and (d) Shanghai dataset.

18 192 × 18 164 × 1 pixels. Among these classes, the GT10 has the largest number of train and test samples, accounting for 14 055 train samples and 581 470 test samples.

3) Beijing Dataset: As shown in Fig. 8(c), the Beijing dataset contains a total of ten annotated classes (from GT1 to GT10, they are as follows: Water, Road, Bare Land, Low Vegetation, Blue Building, High Vegetation, Dense High Building, Dense Low Building, Red Building, and White Building). The MS image is a color image with 6908 × 7300 × 4 pixels. The PAN image contains 27 632 × 29 200 × 1 pixels. Among these classes, the GT7 has the largest number of train and test samples, accounting for 99 735 train samples and 3 347 684 test samples.

4) Shanghai Dataset: As shown in Fig. 8(d), the Shanghai dataset contains a total of 10 manually annotated classes (from GT1 to GT10, they are as follows: Water, Road, Bare Land, Low Vegetation, Blue Building, High Vegetation, Dense High Building, Dense Low Building, Red Building, and White Building). The MS image is a color image with 6908 × 7300 × 4 pixels. The PAN image contains 27 632 × 29 200 × 1 pixels. Among these classes, the GT7 has the largest number of train and test samples, accounting for 173 754 train samples and 2 706 035 test samples.

B. Experiment Settings and Environment

We divided the dataset into different patches to input into the network. For each pixel in the MS data, we extracted a patch of size 16 × 16 × 4 centered around the pixel. Since the spatial resolution of the PAN data is four times higher than the MS data, to ensure spatial correspondence between the patches, we extracted patches of size 64 × 64 × 1 centered around corresponding pixels in the PAN data. The training data and the testing data have no pixel overlap whatsoever. We selected three metrics to evaluate our proposed method:

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TABLE II
EXPERIMENTAL RESULTS WITH DIFFERENT BLUR FACTOR

σ	dataset	1.2	1.4	1.6	1.8	2.0
OA(%)	Hohhot	73.25	73.98	**74.80**	73.69	72.84
	Xi'an	79.83	79.57	79.79	80.01	**80.34**
	Beijing	70.24	71.07	**71.61**	70.73	69.31
	Shanghai	**67.72**	67.15	67.62	67.15	67.49

TABLE III
EXPERIMENTAL RESULTS WITH DIFFERENT KERNEL SIZE

KERNEL SIZE	dataset	3	5	7
OA(%)	Hohhot	**74.80**	73.35	72.46
	Xi'an	**79.79**	79.67	79.19
	Beijing	**71.61**	70.13	69.62
	Shanghai	67.62	**68.52**	66.76

Overall Accuracy (OA), Average Accuracy (AA), and Kappa statistics (KAPPA).

During the training process, we used a batch size of 256 for 30 epochs, with a fixed learning rate of 0.001. To ensure the robustness of our results, we conducted five experiments on datasets and averaged the results to obtain the final performance evaluation of our method. All experiments were ran on the system with RTX4090 24-GB GPU and 128-GB RAM under ubuntu 20.04 LTS, using the PyTorch framework.

C. Hyperparameters Analysis and Selection

In this section, we use the method of controlling variables to investigate the impact of different hyperparameter values on the classification results.

1) Gaussian Blur Factor σ_0: As shown in (5), σ_0 influences the blurriness level of continuous-scale images. Using a fixed Gaussian blur kernel size of 3, experiments were conducted at intervals of 0.2 from 1.2 to 2.0, and the results on four datasets are shown in Table II. At $\sigma_0 = 1.6$, the OA of Hohhot and Beijing datasets reaches the highest level, and subsequently decreases with further increases in σ_0. The results on the Xi'an dataset steadily increase from 1.4 to 2.0, but at σ_0 equals 1.2, its OA is higher than at σ_0 equals 1.4 and 1.6. The results on the Shanghai dataset, however, oscillate consistently without a clear trend. The highest OA is observed at σ_0 equals 1.2.

2) Kernel Size of Gaussian Blur: This parameter corresponds to the size of the Gaussian blur kernel in (13) and (14). With a fixed value of $\sigma_0 = 1.6$, we conducted experiments using kernel sizes of 3, 5, and 7. As shown in Table III, when the kernel size is 3, the testing OA is the highest for the Hohhot, Xi'an, and Beijing datasets. However, for the Shanghai dataset, the testing OA reaching the highest when the kernel size is 5.

D. Ablation Study

We conducted ablation experiments on our proposed strategy using the Hohhot dataset. The backbone we used was 3-D ResNet18, and when the TCIG was removed, we could only use the 2-D network to extract features, so we used a dual-branch ResNet18 as the baseline. We performed ablation experiments on TCIG by dividing it into two strategies: continuous-scale input (CSI) and TE, in conjunction with CMKI. The experimental results are shown in Table IV.

TABLE IV
EXPERIMENTAL RESULTS OF ABLATION STUDY ON HOHHOT DATASET

Baseline	CSI	TE	CMKI	OA(%)	AA(%)	KAPPA(%)
✓				65.60	69.35	62.88
✓	✓			71.32	74.79	67.61
✓	✓	✓		73.13	76.76	69.46
✓	✓		✓	74.05	77.12	70.71
✓	✓	✓	✓	**74.80**	**77.94**	**71.56**

Compared to the baseline, CSI improved the experimental OA by 5.72%. This demonstrates that our proposed CSI strategy and the approach of 3-D convolution can enhance the network's feature representation capability, thereby improving classification accuracy. Building upon CSI, the experimental results with the addition of TE and CMKI also show improvements. However, the experimental results with the addition of CMKI exhibit a greater improvement compared to adding TE. This is because TE tends to be more of a feature enhancement method, while CMKI is a feature representation strategy that aligns better with the task, thus enhancing the overall feature representation capability of the network. When these three strategies are combined, they achieve the highest classification accuracy. We performed a T-test on the results of the five experiments under different ablation conditions versus the results under the intact conditions, and the p-values obtained are all less than 0.05, indicating that these results are statistically significant.

E. Comparison With Other Methods

In this section, we will compare our proposed SNCF-Net with other SOTA methods on the four datasets to validate its effectiveness. We will perform the comparison with NNCNet [36], MFT [18], CRHFF [37], CALC [38], and GCFNet [39] using the same parameter settings as described in their original papers to achieve their best performance.

By conducting this comparative analysis, we aim to demonstrate the superior performance of our SNCF-Net and showcase its potential in remote sensing image classification tasks. The evaluation will be based on various performance metrics, such as OA, AA, and KAPPA, to ensure a comprehensive assessment of each method's performance.

1) Experimental Results on the Hohhot Dataset: The quantitative results of different methods on the Hohhot dataset are shown in Table V. The qualitative experimental results on the Hohhot dataset are shown in Fig. 9. Then, we analyze their experimental results.

NNCNet improves the classification performance of multisource remote sensing data through neighborhood-based data augmentation and feature interaction modules based on linear attention. Compared to our method, it performs better on the GT4: Vegetation and GT5: Intensive Building. This is because these two classes appear as contiguous regions in the dataset. NNCNet's neighborhood-based data augmentation strategy can suppress pixels in regions that have features different from the region's class, achieving better regional segmentation performance. On the other hand, our method

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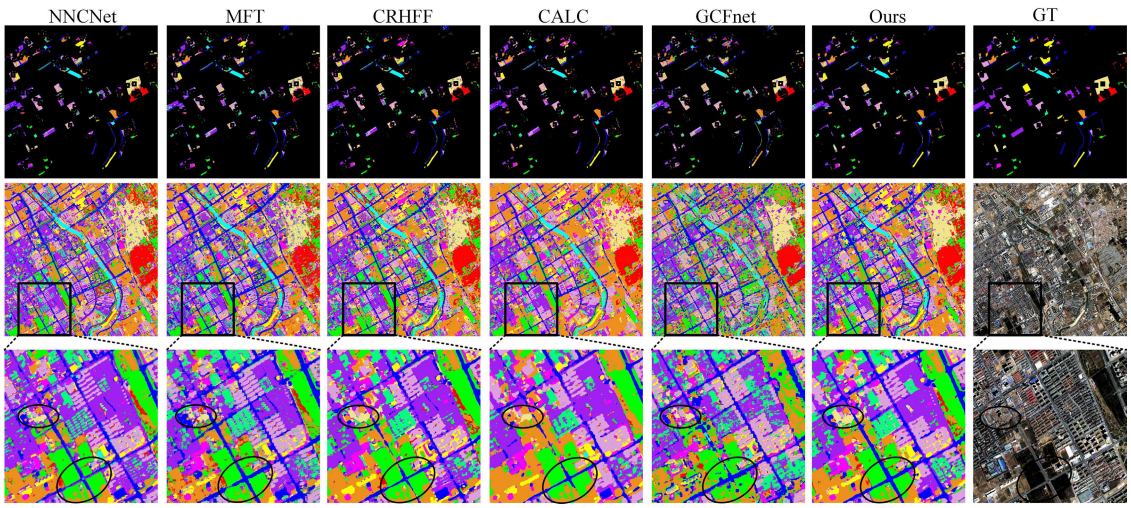


Fig. 9. Classification maps about the test set and overall image on Hohhot dataset. The first row is the classification maps of the test set, the second row is the classification maps of the overall image, and the last row is an enlargement of a portion of the image in the second row. The last column corresponds to the GT map and MS visualization.

TABLE V
QUANTITATIVE RESULTS ON HOHHOT DATASET

Models	NNCNet	MFT	CRHFF	CALC	GCFNet	OURS
c_1	87.55	96.18	95.8	**96.91**	86.7	95.21
c_2	75.4	76.46	81.48	70.45	62.2	**84.61**
c_3	84.76	81.99	88.48	**95.02**	68.84	87.92
c_4	85.24	69.03	**87.85**	82.52	76.07	80.81
c_5	**67.63**	54.23	59.57	65.52	58.99	59.59
c_6	60.41	60.83	60.56	**70.69**	61.91	65.43
c_7	73.94	67.83	69.98	67.15	70.1	**77.31**
c_8	91.94	91.36	**93.53**	92.84	89.51	93.37
c_9	52.93	58.79	55.16	50.88	54.46	**66.37**
c_{10}	57.38	**82.46**	78.5	67.87	32.37	73.16
c_{11}	59.86	75.04	**77.11**	72.91	32.03	73.61
OA(%)	70.85	69.33	72.21	72.5	63.77	**74.80**
AA(%)	72.46	74.02	77.09	75.7	63.01	**77.94**
KAPPA(%)	67	65.47	68.82	69.02	58.99	**71.56**
Times(s)	83.42	15.62	150.86	25.25	265.81	296.37
Params	2.22M	0.21M	4.91M	0.11M	3.89M	6.50M

excels in extracting features from all ground objects, which may result in misclassifying some pixels in certain regions with features different from the class. Our method outperforms NNCNet on the remaining classes and three evaluation metrics, demonstrating its powerful feature extraction capabilities.

MFT is a transformer-based multisource remote sensing image classification method. It pays more attention to data with rich spectral information, while other types of data are embedded into the network after tokenization. It performs better than our method on classes GT1: Water, GT10: Red Building and GT11: Blue Building, which have more prominent spectral information. However, our method significantly outperforms MFT on the remaining classes and evaluation metrics.

CRHFF is a method that fuses cross-resolution features for classification. It can extract features of different ground objects at three scales. Compared to our method, the differences in performance are not significant in four classes: GT1, GT3, GT5, and GT8. However, our method exhibits significant advantages in four classes among the remaining seven, and our OA, AA, and KAPPA are all higher than those of CRHFF. This is due to our feature extraction across continuous-scales, which is superior to CRHFF's three scales, and our method reduces the redundancy of deep semantic features.

CALC fuses features from three scales for remote sensing image classification while focusing on extracting shared and unique features. Compared to our method, the differences in performance are not significant in four classes: GT1, GT5, GT8, and GT11. Similarly, among the remaining seven classes, our method exhibits significant advantages in four classes, and our OA, AA, and KAPPA are all higher than those of CALC. Such comparison results are quite similar to those of CRHFF just discussed. This also indicates the effectiveness of our strategy to extract features across continuous scales.

GCFNet is a patch-free method for PAN and MS classification. Compared to our method, its classification accuracy and evaluation metrics for all classes are significantly lower. This is because the datasets we use have larger size and more complex ground objects, which GCFNet does not adequately consider, making it difficult to competitively model. Overall, due to our comprehensive consideration of PAN and MS characteristics and the powerful feature extraction capability of SNCF-Net, our method significantly outperforms these comparative algorithms in the Hohhot dataset for classification performance.

2) Experimental Results on the Xi'an Dataset: The experimental results on the Xi'an dataset are shown in Table VI and Fig. 10. Our proposed SNCF-Net still achieved excellent results on this dataset. We achieved the highest classification accuracy on GT3: Road2, GT4: Bare Land, GT6: Farmland, GT10: Suburb Building and GT12: White Building. Due to the stable and reliable feature extraction capability of our proposed method, we also achieve consistent performance in other classes. Our method also achieves the highest OA, AA, and KAPPA on the Xi'an dataset.

3) Experimental Results on the Beijing Dataset: As shown in Table VII and Fig. 11, our method achieves the highest accuracy only in GT2: Road and GT3: Bare Land on the

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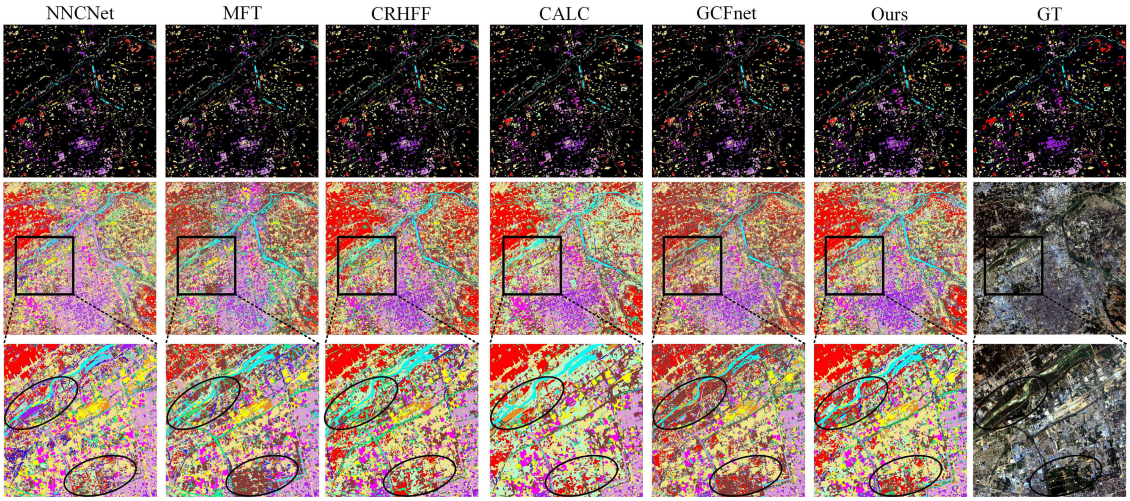


Fig. 10. Classification maps about the test set and overall image on Xi'an dataset. The first row is the classification maps of the test set, the second row is the classification maps of the overall image, and the last row is an enlargement of a portion of the image in the second row. The last column corresponds to the GT map and MS visualization.

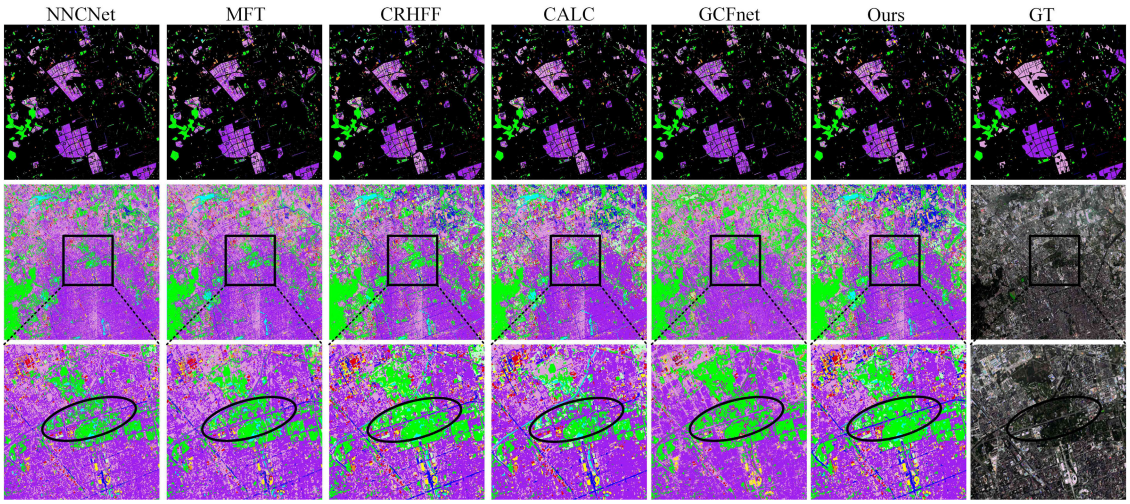


Fig. 11. Classification maps about the test set and overall image on Beijing dataset. The first row is the classification maps of the test set, the second row is the classification maps of the overall image, and the last row is an enlargement of a portion of the image in the second row. The last column corresponds to the GT map and MS visualization.

TABLE VI

QUANTITATIVE RESULTS ON XI'AN DATASET

Models	NNCNet	MFT	CRHFF	CALC	GCFNet	OURS
c_1	93.15	97.35	95.89	**97.59**	95.89	96.90
c_2	**34.16**	16.82	18.35	12.71	17.46	26.84
c_3	62.14	50.69	62.01	42.79	55.38	**62.17**
c_4	69.01	64.58	66.45	39.02	66.46	**69.83**
c_5	55.1	46.68	55.51	**74.46**	36.06	45.09
c_6	68.41	51.69	69.74	65.03	64.31	**75.11**
c_7	77.92	**91.78**	84.84	80.48	89.12	86.27
c_8	74.62	62.85	**77.59**	70.91	72.65	77.09
c_9	77.48	75.73	77.18	**82.01**	76.4	76.63
c_{10}	96.81	91.9	95.51	96.93	97.81	**97.88**
c_{11}	94.23	94.52	95.39	**96.74**	90.71	94.56
c_{12}	76.02	72.87	75.33	74.14	66.36	**76.38**
OA(%)	79.24	74.65	79.42	78.24	76.86	**80.34**
AA(%)	73.25	68.12	72.81	69.4	69.05	**73.73**
KAPPA(%)	76.24	71.08	76.44	75.11	73.48	**77.51**
Times(s)	756.16	140.23	1159.44	199.83	2474.43	2595.99
Params	2.22M	0.21M	4.91M	0.11M	3.89M	6.50M

TABLE VII

QUANTITATIVE RESULTS ON BEIJING DATASET

Models	NNCNet	MFT	CRHFF	CALC	GCFNet	OURS
c_1	64.89	62.39	69.17	**76.39**	2.09	68.06
c_2	14.59	9.62	25.98	22.6	0.51	**27.05**
c_3	64.58	53.5	61.22	62.38	54.49	**72.89**
c_4	72.32	71.37	86.55	**91.45**	51.73	87.89
c_5	66.52	71.83	77.24	**80.29**	45.01	80.15
c_6	90.89	92.3	**95.01**	90.11	90.75	94.74
c_7	69.83	54.35	67.24	**71.83**	66.28	66.85
c_8	56.97	**69.03**	57.24	60.39	44.72	65.18
c_9	67.97	57.64	75.27	**83.65**	44.3	83.29
c_{10}	79.27	76.73	**88.64**	87	50	87.11
OA(%)	68.69	65.27	69.05	71.4	61.47	**71.61**
AA(%)	64.78	61.88	70.36	72.61	44.99	**73.32**
KAPPA(%)	55.09	51.27	56.68	59.13	44.15	**60.84**
Times(s)	2741.95	536.3	4952.43	2370.07	7736.46	8065.63
Params	2.22M	0.21M	4.91M	0.11M	3.89M	6.50M

Beijing dataset, while CALC achieves the highest accuracy in GT1: Water, GT4: Low Vegetation, GT5: Blue Building, GT7: Dense High Building and GT9: Red Building. However, our method's OA, AA, and KAPPA are still higher than those of CALC. This is because our method maintains

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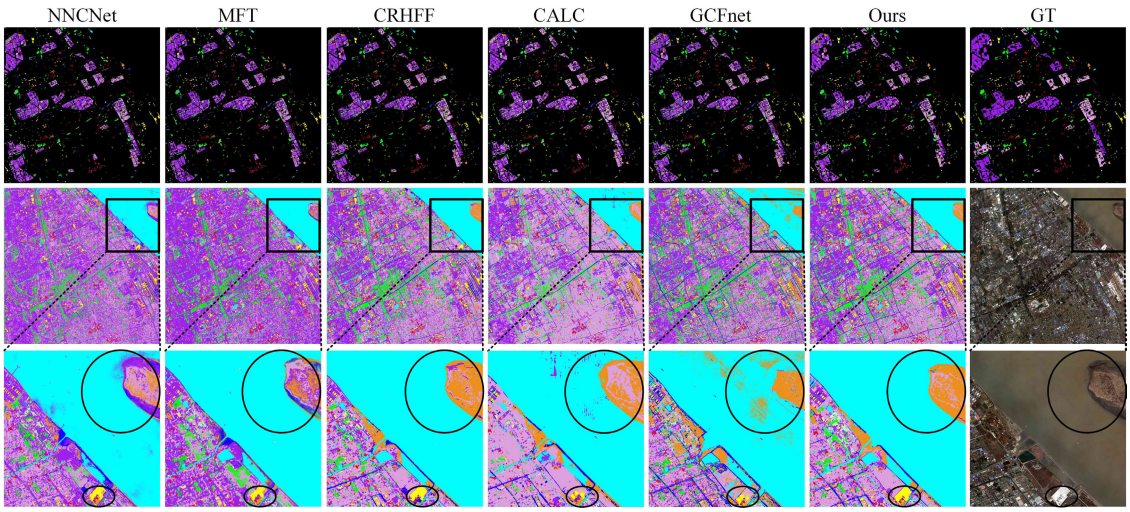


Fig. 12. Classification maps about the test set and overall image on Shanghai dataset. The first row is the classification maps of the test set, the second row is the classification maps of the overall image, and the last row is an enlargement of a portion of the image in the second row. The last column corresponds to the GT map and MS visualization.

TABLE VIII
QUANTITATIVE RESULTS ON SHANGHAI DATASET

Models	NNCNet	MFT	CRHFF	CALC	GCFNet	OURS
c_1	73.74	73.37	81.87	75.28	71.34	**82.28**
c_2	42.24	17.57	44.86	40.33	41.16	**48.57**
c_3	81.86	73.71	80.49	84.08	83.16	**84.51**
c_4	80.23	61.44	79.53	75.55	69.92	**81.71**
c_5	75.75	70.15	84.95	74.01	83.82	**88.83**
c_6	80.85	79.09	84.19	59.42	82.26	**85.95**
c_7	70.32	**73.83**	69.55	59.14	64.22	70.72
c_8	47.7	39.75	50.47	**64.34**	45.87	49.60
c_9	36.4	**53.42**	52.47	29.47	10.20	49.94
c_{10}	96.15	92.77	97.06	85.9	93.49	**99.40**
OA(%)	66.42	65.29	67.6	62.24	62.66	**68.52**
AA(%)	68.52	63.51	72.54	64.75	64.54	**74.15**
KAPPA(%)	49.37	46.37	52	44.76	43.35	**53.31**
Times(s)	1723.11	308.97	2477	562.98	2727.74	3528.06
Params	2.22M	0.21M	4.91M	0.11M	3.89M	6.50M

competitive accuracy across all classes, while achieving significantly higher accuracy in the second most abundant class, GT8: Dense Low Building, and the third most abundant class, GT6: High Vegetation, compared to CALC. Although the margin of superiority over CALC is small, we still achieve the highest OA, AA, and KAPPA on the Beijing dataset.

4) Experimental Results on the Shanghai Dataset: The experimental results on the Shanghai dataset are shown in Table VIII and Fig. 12. MFT achieves the highest accuracy on GT7: Dense High Building and GT9: Red Building, while CALC achieves the highest accuracy on GT8: Dense Low Building. For the remaining classes, our method achieves the highest accuracy. Nevertheless, our method still achieves the highest OA, AA, and KAPPA.

V. CONCLUSION

In this article, we introduce a SNCF-Net for PAN and MS classification and achieve competitive results on four datasets. Experimental results demonstrate that SNCF-Net can adaptively select the appropriate scale within the texture-enhanced continuous-scale images generated by TCIG to extract features of different land cover types. This provides a fresh perspective for multiscale feature extraction in remote sensing imagery. The CMKI module effectively reduces redundant representations in deep features of PAN and MS, contributing to improved classification accuracy. However, when the TCIG module generates continuous-scale images, the computational cost increases sharply with the increase in the size of the input images and the number of different scale images produced. Selecting suitable parameters for different tasks requires some time and effort, and there inevitably exists unnecessary redundancy between these continuous-scale images. In the future, we will explore a learnable approach for generating continuous-scale images while considering redundancy between continuous scales, achieving an adaptive and sparse continuous-scale feature extraction method.

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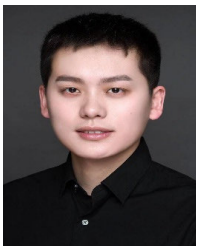
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